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Study and Research Project

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COLLABORATIVE EDITING AND VISUALIZATION OF MAPS IN HETEROGENEOUS VIRTUAL ENVIRONMENTS

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Contents

Contents 3

1 Introduction 5
- 1.1 Abstract . . . 5
- 1.2 Context and Motivation . . . 5
- 1.3 Problem Statement . . . 6
- 1.4 Objectives, Scope, and Personal Contribution . . . 6

2 State of the Art 7
- 2.1 Chapter Overview . . . 7
- 2.2 Geospatial Fidelity and DEM Comparisons . . . 7
  - 2.2.1 Controlled Comparison: The White Cliffs of Dover . . . 8
  - 2.2.2 Criticality for Professional VR Applications . . . 8
- 2.3 Challenges in Multi-User Immersive Collaboration . . . 8
- 2.4 Taxonomy of Virtual Locomotion . . . 9
  - 2.4.1 Physical and Simulated Gait . . . 9
  - 2.4.2 Steering, Selection, and Manipulation . . . 9
  - 2.4.3 Suitability for Geospatial Environments . . . 10
- 2.5 Architecture: Centralization vs. Distribution . . . 10
- 2.6 State of the Art: Geospatial Usage Cases . . . 10
  - 2.6.1 Collaborative Environmental Simulation . . . 10
  - 2.6.2 Real-Time Emergency Response and Management . . . 10
  - 2.6.3 Collaborative Decision Support in "Data Worlds" . . . 11

3 System Architecture and Interaction Design 13
- 3.1 Network Architecture and Host-Client Synchronization . . . 13
  - 3.1.1 Host Authority and Connection Flow . . . 13
  - 3.1.2 Development Methodology . . . 13
  - 3.1.3 Bandwidth and Architectural Constraints . . . 14
- 3.2 Immersive Analytics Ecosystem and Component Choices . . . 14
- 3.3 The Design-Render-Display Pipeline . . . 15
- 3.4 Unified Interaction Logic and Input Mapping . . . 15
  - 3.4.1 Contextual Binding Strategy . . . 16
  - 3.4.2 Design Intent and Usability . . . 16
  - 3.4.3 Technical Implementation: XR Raycasting . . . 16
- 3.5 Collaborative Annotation Tools . . . 16
  - 3.5.1 Marker Placement and Terrain Alignment . . . 16
  - 3.5.2 Procedural Zone Generation . . . 17
- 3.6 Multi-Modal Navigation and Comfort Mechanics . . . 17
- 3.7 HMI and Awareness Cues . . . 17
  - 3.7.1 Head-Coupled Interface . . . 17

3.7.2 Avatar Representation and Limitations 17
3.8 Prototype Maturity and Functional Scope 18

4 Discussion and Conclusion **21**
4.1 Discussion and Analysis of Results 21
4.1.1 Interaction and Leadership 21
4.1.2 Evolution of the Project Scope 21
4.2 Technical Limitations 21
4.3 Conclusion and Future Perspectives 22
4.3.1 Future Progress 22

5 Bibliography **25**

Chapter 1

Introduction

1.1 Abstract

The integration of Geographic Information Systems (GIS) with Virtual Reality (VR) has initiated a paradigm shift in spatial planning, evolving from static 2D maps to interactive 3D "Digital Earths" [3]. However, deploying these systems for collaborative professional use introduces significant challenges, particularly when users operate on heterogeneous hardware. This report details the design and implementation of a Collaborative Immersive Analytics prototype using Unity, Cesium for Unity, and Netcode for GameObjects. The project addresses three core scientific barriers: network synchronization of spatial annotations, human-machine interface (HMI) design for awareness cues, and the mitigation of cybersickness during world-scale virtual locomotion.

1.2 Context and Motivation

Historically, cartography and spatial planning used two dimensional static representations. However the rise of "Digital Earth" technologies and Virtual Reality has transformed the interaction with geospatial data. A Digital Earth is defined as a comprehensive, multi-resolution, three-dimensional virtual representation of the planet that integrates massive amounts of geographic, topographical, and infrastructural data [3]. Today, immersive 3D visualization is a major asset for domains such as urban-planning, geo-ecology, risk assessment and emergency responses [9].

Technologies like Cesium for Unity have become foundational in this domain, enabling the dynamic streaming and rendering of world-scale 3D tiles and photogrammetry directly into real-time graphical engines [7].

Concurrently, the demand for cooperative cartography has grown. Present day decision making infrequently happens in solitariness: numerous specialists must work together to synthesize complex spatial variables. As Hackathorn and Margolis (2016) note, the ambition of immersive analytics is to make 'virtual data worlds' that blend analytical reasoning with immersive spaces. By converting the Digital Earth into a persistent, social, and engaging environment, teams can achieve a shared "sense of presence" allowing urban planners, ecologists, and emergency responders to collaboratively interact with multi-sensory data in real time [4].

1.3 Problem Statement

Developing a collaborative geospatial VR system introduces several major hurdles: in heterogeneous collaborative virtual environments, participants often interact using different hardware, such as Head-Mounted Displays (HMDs) or standard web browsers.

This system needs robust awareness cues: visual indicators like gaze vectors and 3D pointers to maintain a sense of social presence and shared focus. It requires data synchronization: maintaining state consistency for map modifications (e.g., drawing zones) across a network is critical. Without a centralized authority, concurrent edits can lead to divergent states between users. Virtual locomotion and cybersickness; navigating a 1:1 scale globe triggers cybersickness, a physiological response to the conflict between perceived visual motion and physical stillness. The immense scale of geospatial data necessitates locomotion techniques that allow rapid movement without inducing nausea.

1.4 Objectives, Scope, and Personal Contribution

The objective of this project was not to develop a complete production-ready collaborative GIS platform, but rather to build a functional research prototype demonstrating how collaborative map annotation can be implemented in immersive VR. The prototype relies on existing frameworks, mainly Unity, Cesium for Unity, the XR Interaction Toolkit, and Netcode for GameObjects. My contribution consisted in integrating these components into a coherent collaborative workflow and implementing the interaction and synchronization mechanisms required for VR-based geospatial annotation.

Multi-user distributed architecture: I implemented a robust host-client networking architecture using Unity's Netcode for GameObjects to enable real-time synchronization of customized map annotations. Rather than relying on a dedicated standalone server, the system designates one participant as the host, who serves as the authoritative entity responsible for maintaining the global state of persistent markers and procedurally defined polygonal zones. This approach follows a *centralized management, distributed application* paradigm, with the objective of maintaining a consistent shared state across connected users.

Comfort-oriented locomotion paradigms: I developed a multi-mode navigation system designed to enhance user comfort in virtual reality. This system combines selection-based teleportation widely recognized for reducing cybersickness with a top-down view interaction mode. The top-down perspective allows users to traverse large distances using continuous, flight like motion at high altitude, thereby minimizing sensory conflicts and excessive optical flow typically associated with ground-level continuous locomotion. Additionally, a flying mode based on left joystick input and gaze direction enables more precise exploration and interaction.

Ergonomic Human-Machine Interface (HMI): I designed a contextual follow-head menu that acts as a dynamic interaction interface for VR-based map systems. The interface continuously repositions and reorients itself within the user's peripheral field of view, ensuring that tools for creating and managing markers and zones remain accessible without obstructing the immersive 360-degree environment. This spatial user interface leverages tracked-device raycasting to provide an intuitive, gesture-based interaction experience.

Chapter 2

State of the Art

2.1 Chapter Overview

This chapter presents the scientific and technical background required to understand the design choices of the prototype. It is structured around four complementary aspects.

First, geospatial fidelity is discussed because the relevance of a virtual map strongly depends on the accuracy of the underlying terrain model. This justifies the use of high-resolution 3D terrain data, especially for professional applications such as urban planning or emergency response.

Second, the chapter reviews the main challenges of multi-user immersive collaboration. Since the project aims to support several users working on the same geospatial scene, issues such as awareness cues, social presence, latency, and shared visual references are essential.

Third, virtual locomotion techniques are introduced in order to justify the navigation choices made in the prototype. In world-scale environments, movement is not only a technical problem but also a comfort issue, because inappropriate locomotion methods can easily induce cybersickness.

Finally, the chapter discusses architectural choices for collaborative systems, especially the opposition between distributed and centralized models. This provides the theoretical basis for the host-authority architecture used in the implementation.

2.2 Geospatial Fidelity and DEM Comparisons

A core requirement of geospatial systems is terrain accuracy, which is largely determined by the resolution of the Digital Elevation Model (DEM). Zhu et al. (2019), in *Multiple-view geospatial comparison using web-based virtual globes* [10], demonstrate that low-resolution models, such as 90 m grids, are often overly smoothed and fail to accurately represent real-world terrain. In a controlled study of the White Cliffs of Dover, a 90 m resolution model exhibited vertical errors of approximately 45 m.

In contrast, Cesium World Terrain, which provides a resolution of 1 m, enables a high-fidelity representation of real-world geometry. Such accuracy is essential for professional applications.

2.2.1 Controlled Comparison: The White Cliffs of Dover

Research by Zhu et al. (2019) [10] employed a controlled experimental setup to evaluate the impact of terrain resolution by visualizing the White Cliffs of Dover (see Fig. 2.1). In this study, both the imagery and reference points were kept constant, while the underlying Digital Elevation Model (DEM) was varied across three levels: VR-TheWorld (90 m), a legacy terrain service in which the coarse resolution results in significant smoothing and vertical altitude errors of approximately 45 m; STK World Terrain (30 m), a model provided by Systems Tool Kit (STK), an industry-standard software suite used in aerospace and defense [5], which offers improved precision but still represents an approximation of real world geometry; and Cesium World Terrain (up to 1 m), a high-fidelity global terrain service that most accurately represents the real geometry of the cliffs, providing a close match to satellite imagery and ground-truth elevation data.

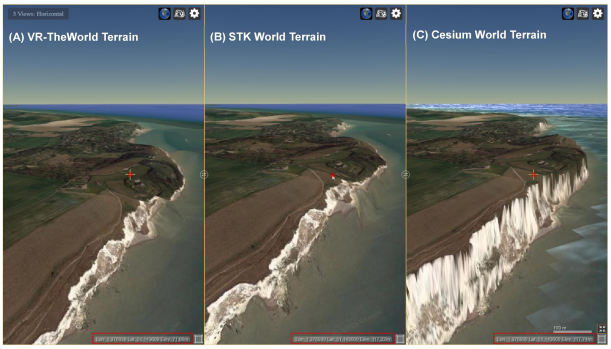


Figure 2.1: Visualization of the White Cliffs of Dover used for comparing terrain representations across different DEM resolutions [10].

2.2.2 Criticality for Professional VR Applications

Precision in Decision Support: for use cases such as urban planning or emergency response (e.g., landslide management), sub-meter precision is required to identify safe building sites or intervention routes. Using high-resolution models ensures that the "data world" is a reliable replica of the physical environment, which is essential for accurate analytical reasoning.

2.3 Challenges in Multi-User Immersive Collaboration

Collaborating in VR is hindered by communication barriers and technical usability issues. The lack of nonverbal cues (body language, facial expressions) requires the creation of shared visual goals and awareness cues to artificially reconstruct social presence. Furthermore, immersive environments demand real-time, low-latency communication; hardware latency and tracking inaccuracies can lead to inconsistent feedback, breaking the collaborative experience.

To address these challenges, several solutions are required:

- **Networking infrastructure:** utilizing cloud-based architectures and future 6G standards to ensure high Quality of Service (QoS) and minimal latency.
- **Awareness cue reconstruction:** implementing realistic avatars and visible gaze/pointing vectors to provide social cues.

- **Usability and leadership:** establishing shared visual goals and defining facilitator or "leader" roles to guide participants through complex virtual tasks.
- **Interface consistency:** developing intuitive navigation and consistent control schemes to assist users with varying technical skill levels [2].

2.4 Taxonomy of Virtual Locomotion

Navigating within an immersive environment, particularly one as vast as a 3D digital globe, requires a structured approach to movement. According to *Virtual Locomotion: A Survey* [1], locomotion techniques can be classified into six primary categories, each presenting distinct trade-offs in terms of usability, performance, and user comfort.

2.4.1 Physical and Simulated Gait

- **Full gait:** these techniques are based on real walking, in which the user performs a complete walking cycle. This includes physical walking within a tracked space or redirected walking, where the virtual scene is subtly rotated to keep the user within physical boundaries. While these solutions are highly natural and provide excellent spatial perception, they require significant physical space, making them difficult to implement in large-scale virtual environments.
- **Partial gait:** these methods rely on a partial reproduction of the walking cycle, such as walking-in-place. The system translates the user's vertical motion into virtual forward movement. This approach reduces physical space constraints but is less natural than full gait and can become fatiguing over extended use.
- **Gait negation:** these techniques aim to preserve a full walking experience while keeping the user stationary through specialized hardware, such as omnidirectional treadmills or mechanical platforms. Although they offer high realism, they remain largely inaccessible due to the cost and bulk of the required equipment.

2.4.2 Steering, Selection, and Manipulation

- **Steering:** this category allows continuous control of movement via a joystick, gaze direction, or hand gestures. Steering is highly effective for traversing large-scale environments; however, it is a primary contributor to cybersickness, as continuous visual motion conflicts with the user's stationary vestibular system.
- **Selection-based (teleportation):** as detailed in *Teleportation in Virtual Reality: A Mini-Review* [6], selection-based techniques allow users to move instantly to a chosen destination. This method significantly reduces the risk of cybersickness and is highly efficient for covering long distances, although the discontinuous nature of the movement can occasionally impair spatial orientation.
- **Manipulation-based:** these techniques involve manipulating the virtual environment itself, for example, by "grabbing" the terrain to pull oneself forward or scaling the entire world. They are particularly well suited to world-scale geospatial data, as they allow users to manage scale without physical constraints while maintaining a high level of comfort.

2.4.3 Suitability for Geospatial Environments

In the specific context of an immersive cartography system where distances are immense, selection-based (teleportation) and manipulation-based approaches are significantly more adapted than gait-based techniques. By providing a safe and efficient means to traverse a 3D globe, these methods allow the user to focus on analytical reasoning and collaboration rather than the physical limitations of the workspace.

2.5 Architecture: Centralization vs. Distribution

Modern governmental and professional information architectures are undergoing a clear evolution, shifting from distributed "net" topologies which frequently suffer from inefficient resource duplication toward centralized "star" topologies facilitated by robust cloud computing infrastructure. This centralized model is fundamentally defined by the principle of "centralized management, distributed application," creating an ideal environment for advanced Geographic Information Systems (GIS).

The adoption of a star structure enables high-performance computing capabilities, integrated data management, and agile cross-departmental business collaboration. By relying on a centralized authoritative core, the system allows diverse distributed clients ranging from immersive Head-Mounted Displays (HMDs) to traditional desktop monitors to share and modify high-fidelity geospatial resources effectively within a unified collaborative environment [8].

2.6 State of the Art: Geospatial Usage Cases

The practical application of immersive geospatial technologies is rapidly expanding across several professional sectors, moving from simple visualization to "Immersive Analytics" in shared data worlds.

2.6.1 Collaborative Environmental Simulation

As explored in *GeospatialVR: A web-based virtual reality framework for collaborative environmental simulations* [7], web-based VR frameworks are being utilized to democratize access to complex environmental models. This system allows distributed teams to collaboratively engage with flood simulations and ecological data in real-time, bridging the gap between traditional GIS and immersive simulation.

2.6.2 Real-Time Emergency Response and Management

In the research concerning 3D emergency response systems using Cesium Digital Earth, high-fidelity virtual globes are used for landslide disaster management [9]. These systems integrate real-time data, including:

- Drone video feeds and sensor information.
- Live tracking of emergency response teams.
- 3D site reconstruction for command-level spatial perception.

2.6.3 Collaborative Decision Support in "Data Worlds"

Modern systems utilize the framework of "Immersive Analytics" to build virtual data worlds. These environments support both exploratory and predictive analytics, allowing experts to inhabit the data and reach consensus on complex infrastructural or ecological projects in a shared, multi-sensory environment [4].

Chapter 3

System Architecture and Interaction Design

3.1 Network Architecture and Host-Client Synchronization

Following the "centralized management, distributed application" paradigm, the system utilizes a **star topology** managed by a central Unity host [8]. Due to the absence of a dedicated server infrastructure for this project, the architecture relies on a **host-client model** to facilitate multi-user synchronization.

3.1.1 Host Authority and Connection Flow

The `NetworkStartUI.cs` script manages the initial handshake and session initialization. While the system is technically capable of supporting a dedicated server, the current implementation is configured for **host mode**, where one participant's machine functions as the authoritative server while simultaneously acting as a local client. The host manages the unified **data world state**, validating all user interactions before replicating them to connected peers.

3.1.2 Development Methodology

The implementation followed a progressive prototyping approach. At the beginning of the project, the first networking tests were carried out on PC in order to understand and validate the basic behavior of Netcode for GameObjects, such as player spawning, host-client connection, and networked object instantiation. These early tests were not intended to represent the final interface of the application. They mainly served as a draft environment to experiment with the first ideas of collaborative interaction without the additional complexity of VR input, headset constraints, and immersive user interface design.

After this initial PC-based prototyping phase, the implementation was progressively adapted to VR. The focus then shifted toward controller-based raycasting, the contextual VR menu, headset-oriented interaction, marker and zone placement from the VR controller, and comfort-oriented navigation modes. This development process made it possible to separate the validation of the core networking principles from the later design of the immersive interaction workflow.

3.1.3 Bandwidth and Architectural Constraints

The reliance on a host-client structure introduces specific constraints regarding network throughput:

- **Data burden:** the host machine carries the dual responsibility of rendering the high-fidelity *Cesium* geometry while broadcasting state updates to all connected clients.
- **Bandwidth and latency:** without a dedicated server to offload bandwidth management, the host machine must process all network traffic, which can result in synchronization latency as annotation complexity increases.
- **State consistency:** the `disableVRContextMenuUntilConnected` flag prevents clients from modifying data until they have successfully synchronized with the host's authoritative core.

3.2 Immersive Analytics Ecosystem and Component Choices

The selection of technologies for this prototype is guided by the taxonomy of *Immersive Analytics* components. As illustrated in Figure 3.1, the ecosystem is categorized into three primary pillars: rendering engines, analytics platforms, and 3D displays[4].

Rendering Engines	Inside-Out Analytics	3D Displays
Gaming Engines, such as Unity3D, Unreal, CryEngine	**Visual Analytics**, such as Qlik, Tableau	**Head-Mounted Display**, such as Oculus Rift, Samsung GearVR, Google Glasses, Cardboard
HTML5 with WebGL, three.js	**Analytic Platforms**, such as Teradata Aster, Spark/DataBricks, AWS Analytics ML	**Cave Rooms**, such as CAVE2
Online Metaverse, such as High Fidelity, virBELA, OpenSimulator, Second Life	**Analytic Flow IDE**, such as RapidMiner, Alteryx, KMINE	**Desktop** with room projectors
Open Sandbox, such as MineCraft & Lego World	**Custom R and Python Services**	**Input/Feedback Devices**, such as joysticks, spatial sound, tactile gloves

Figure 3.1: Taxonomy of the immersive analytics ecosystem, categorized by rendering capabilities, analytic methods, and display hardware [4].

- **Rendering engine:** this project utilizes **Unity3D**, version 6000.0.38f1, a high-performance gaming engine chosen for its ability to handle the intensive real-time rendering required for world-scale 3D tiles and high-fidelity photogrammetry provided by Cesium [9, 4].
- **XR Interaction Toolkit:** to facilitate complex user interactions, the implementation relies on Unity's **XR Interaction Toolkit**. While the architecture is designed to be hardware-agnostic, the current development and testing focused on the **Meta Quest 3** as the primary HMD (Head-Mounted Display) [2].
- **Heterogeneous potential vs. practical focus:** although the initial project scope envisioned a two-person team managing a wide range of devices, the single-author implementation necessitated a focused effort on VR interaction. However, by utilizing standard XR frameworks, the system maintains compatibility with other VR platforms and

maintains the architectural foundation required for future desktop client integration [4, 8].

3.3 The Design-Render-Display Pipeline

The architecture follows the structured pipeline for building virtual data worlds, facilitating the transition from raw geospatial data to a collaborative environment. This workflow is visualized in figure 3.2.

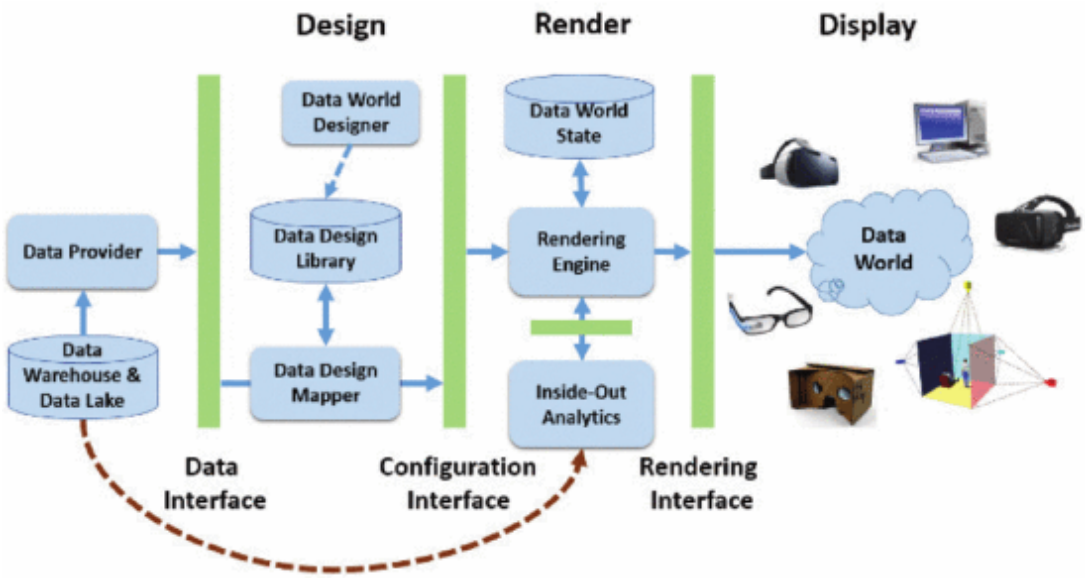


Figure 3.2: The conceptual pipeline for collaborative data worlds, demonstrating the flow from data providers to multi-user displays [4].

- **Design phase:** *Cesium* serves as the primary *Data Provider*, streaming massive topographical data into the Unity environment [9, 10]. The *Data Design Mapper* (Cesium for Unity API) translates global coordinates into local Unity space for rendering.
- **Render phase:** Unity acts as the central *Rendering Engine*, maintaining the *Data World State*. This includes collaborative annotations markers and zones synchronized through `VRToolController.cs` and the Host-authority network model [8].
- **Display phase:** the final *Data World* is projected to the user's interface. While the current implementation prioritizes the immersive experience of the Meta Quest 3, the centralized state management is designed to allow connected clients, regardless of display type, to receive consistent and synchronized geospatial feedback[2, 4].

3.4 Unified Interaction Logic and Input Mapping

A critical design choice in the prototype's Human-Machine Interface (HMI) is the implementation of a **unified interaction pattern**. This approach centralizes all primary actions on a single input binding the **XR Select Action** facilitated by XR Raycasting.

3.4.1 Contextual Binding Strategy

The system utilizes a dual-purpose mapping for the primary selection button (typically the controller trigger or grip). This was a deliberate decision intended to minimize the cognitive load on the user and ensure a consistent control scheme across different modes [2]:

- **Phase 1: tool selection (menu interaction):** when the `VRContextMenu.cs` is active, the `TrackedDeviceGraphicRaycaster` allows the user to point at the interface. By using the select binding, the user interacts with the UI buttons to switch the active state within the `VRToolController.cs` (e.g., changing the tool from "Navigation" to "Add Marker").
- **Phase 2: tool execution (spatial interaction):** Once a tool is selected, the same select binding is repurposed for environmental interaction. When pointing at the *Cesium* terrain, the select action triggers the logic associated with the active `VRToolAction` such as spawning a `NetworkMarker` or dropping a point for a `NetworkZone`.

3.4.2 Design Intent and Usability

This "select to choose, select to use" flow was implemented to address the usability barriers often found in complex immersive environments [2]. By limiting the number of required physical buttons, the system creates a more intuitive "point-and-click" experience similar to traditional 2D interfaces but adapted for 3D space. This consistency is essential for reducing the learning curve, allowing users to focus on the geospatial data rather than mastering a complex controller layout [2].

3.4.3 Technical Implementation: XR Raycasting

The interaction is powered by a continuous **XR Raycast** originating from the controller rig. The system dynamically switches the raycast's target mask:

- When interacting with a **UI layer**, the raycast enables menu navigation through the `TrackedDeviceGraphicRaycaster`.
- When hitting the **Cesium terrain**, the raycast retrieves the exact world-scale coordinates and surface normals (`hit.normal`) required for accurate object placement and synchronization.

3.5 Collaborative Annotation Tools

The core of the Immersive Analytics experience relies on spatial annotations, orchestrated by the `VRToolController.cs` which maps VR inputs to `VRToolAction` enumerations.

3.5.1 Marker Placement and Terrain Alignment

When placing markers, the system performs a physics raycast from the VR controller through the Cesium terrain. To account for the high-resolution geometry of the terrain, the system retrieves the `hit.normal` to orient the marker prefab according to the terrain surface and to slightly offset it from the ground, reducing graphical clipping.

3.5.2 Procedural Zone Generation

The `NetworkZone.cs` script allows users to define areas by dropping sequential points into a synchronized `NetworkList<Vector3>`. Once a zone is closed, the closed state is replicated through the network, and each client locally rebuilds the visual representation of the zone from the synchronized points. The filled mesh is generated by computing a geometric center, which is used as the first vertex, while the boundary points form the remaining vertices. The mesh is then filled using a simple triangle fan structure, where each triangle connects the center to two consecutive boundary points. This method is efficient for simple polygonal zones, but it is not intended to handle complex GIS geometries or highly concave polygons.

3.6 Multi-Modal Navigation and Comfort Mechanics

To balance efficient exploration with the mitigation of cybersickness, the prototype utilizes a hybrid navigation system.

- **Teleportation (selection-based):** for localized movement, the system relies on selection-based teleportation to instantly translate the XR Rig, acting as the primary safeguard against nausea [6, 1].
- **Top-down overview mode:** for rapid traversal over global distances, the user is positioned high above the terrain. By restricting movement to a 2D plane and reducing optical flow, this mode significantly mitigates cybersickness and allows users to view markers and zones from an overhead perspective [6, 1]. This mode is also well suited for placing markers and defining zones.
- **Steering and flight (Continuous):** a continuous movement mode is included for scenarios requiring close-up investigation of specific features, providing high flexibility for exploratory tasks.

3.7 HMI and Awareness Cues

Maintaining presence requires visual indicators of other participants' actions [2].

3.7.1 Head-Coupled Interface

The `VRContextMenu.cs` script implements a UI that remains perpendicular to the user's gaze using `Quaternion.LookRotation`, ensuring tools remain accessible without obstructing the 360-degree geospatial data world.

3.7.2 Avatar Representation and Limitations

The current avatar consists of a sphere for the head and virtual controllers for the hands. While functional, this rig presents limitations such as movement discontinuity during teleportation and unnatural head-tilt angles when a user is in top-down mode looking at the ground. Future iterations could use color-coded states (e.g., **yellow** when adding a marker) to provide immediate contextual awareness.

3.8 Prototype Maturity and Functional Scope

The current system should be understood as a research prototype rather than a complete collaborative GIS application. Its purpose is to validate the feasibility of VR-based collaborative map annotation using a host-client architecture. Some components are functional, while others remain partially validated or exploratory.

Feature	**Maturity level**	**Comments**
Cesium terrain visualization in Unity	Functional	The prototype can display and interact with high-fidelity geospatial terrain using Cesium for Unity.
VR contextual menu	Functional	The head-coupled menu and raycast-based UI interaction were implemented and tested in VR.
Marker placement	Functional prototype	Markers can be placed on the terrain and represented as networked objects. Further work would be required for advanced metadata, editing, and persistence.
Polygonal zone creation	Functional prototype	The user can create simple polygonal zones. However, complex GIS geometry operations, robust editing, and layer management are not fully implemented.
Network synchronization	Partially validated	The host-authority model was first explored through early PC-based Netcode prototypes, mainly to test player spawning, host-client connection, and networked object instantiation. These early PC versions acted as draft implementations before the functionality was adapted to the VR workflow. VR interactions were later tested with one available headset. However, the complete scenario involving several simultaneous HMD users could not be fully evaluated.
Heterogeneous collaboration	Architectural basis only	The architecture was designed with heterogeneous environments in mind, but the current implementation focuses mainly on VR usage with the Meta Quest 3. A complete desktop or web client was not finalized.
Large-scale globe manipulation	Exploratory / not finalized	Attempts were made to support larger scale navigation and city switching, but these features were removed from the final prototype because of persistent bugs.

Table 3.1: Maturity level of the main prototype features.

This maturity analysis is important because it clarifies the scope of the work. The main result of the project is not a finished professional platform, but a functional foundation demonstrating how collaborative annotations, VR interaction, and geospatial visualization can be combined in a shared immersive environment.

Chapter 4

Discussion and Conclusion

4.1 Discussion and Analysis of Results

The current prototype validates some core concepts of collaborative immersive analytics. A significant strength of the system is the ergonomic design of the interface; the simplicity of the menu and the implementation of consistent controls significantly reduce the learning curve for new users.

4.1.1 Interaction and Leadership

While interaction via Raycasting proved effective for this prototype, the architecture allows for the exploration of other direct selection methods in the future. A critical finding for the system's evolution is the potential for an asymmetric **leader** role. A PC-based user could serve as a facilitator, guiding VR participants through the environment, managing data layers, and assisting with complex navigation tasks.

4.1.2 Evolution of the Project Scope

Initially, the project perimeter was broader and was intended to be carried out by two students. It included a wider exploration of heterogeneous collaborative environments, potentially involving different types of clients and additional map data sources such as IGN maps. The final implementation required an adaptation of the scope.

The work therefore shifted toward the most central technical and scientific challenges of the project: *networking*, collaborative annotation tools, *awareness cues*, and ergonomic optimization for HMDs. This narrower scope allowed for a deeper investigation into user comfort, interaction design, and cybersickness mitigation, while leaving the full heterogeneous multi-client platform as a future development direction.

4.2 Technical Limitations

Several limitations were identified during development, primarily stemming from hardware availability and documentation challenges.

- **Hardware and network testing constraints:** during the VR implementation phase, I had access to one headset, which allowed me to develop and test the main VR interactions

directly in immersive conditions. However, access to multiple headsets at the same time was limited, as other available devices were already lent to other students. As a result, the initial Netcode behavior was explored through PC-based prototypes, while the final interaction workflow was developed mainly for VR. The complete collaborative scenario involving several simultaneous HMD users could therefore not be fully evaluated.

- **Framework documentation:** the development process was hindered by the often incomplete or complex documentation of the *XR Interaction Toolkit*. Understanding its nuances required extensive "reverse engineering" and iterative testing of the provided samples.
- **Global Navigation and Manipulation:** efforts to implement a system for switching between different cities were ultimately removed due to persistent bugs. Furthermore, while the current system allows for smooth tool-switching and rotation, a full "atlas-style" world manipulation mode remains a conceptual goal that was not finalized in this version.

4.3 Conclusion and Future Perspectives

This research and development effort has established a foundational prototype for collaborative geospatial mapping in virtual reality. Rather than providing a complete professional platform, the project demonstrates the feasibility of combining Cesium-based geospatial visualization, VR interaction, and host-authority synchronization within a single experimental system.

The project partially addresses the requirements of multi-user synchronization and world-scale VR navigation by integrating networked annotations, a contextual VR interface, and several comfort-oriented locomotion modes into a single prototype. Its main contribution lies in the integration of these mechanisms into a coherent VR workflow, while also identifying the technical limitations that must be solved before such a system can become a mature collaborative tool.

4.3.1 Future Progress

To evolve this prototype into a comprehensive professional tool, the following developments are proposed:

- **Multi-headset and network evaluation:** future testing should involve several simultaneous headset users in order to validate the complete collaborative VR scenario under realistic conditions. This would make it possible to evaluate synchronization behavior, latency, object spawning, annotation replication, and usability when multiple HMD users interact with the same scene at the same time.
- **Improved annotation tools:** future versions should provide more complete annotation features, assigning names to annotations, changing their visual appearance, and improving polygon creation. In particular, zone generation could be improved with more robust triangulation methods in order to support concave or more complex shapes.
- **Multimodal communication:** to better support collaboration, the integration of text, voice chat, or spatialized speech communication would be essential. These features would help users coordinate their actions, explain their intentions, and maintain social presence while working in the shared geospatial environment.

- **Departmental data layering:** implementing layered zones would allow different departments, such as urban planning or emergency services, to visualize and manage distinct data sets within the same shared "data world". This would make the prototype more suitable for professional collaborative decision-making.
- **Network refinement and stress testing:** while the networking layer was implemented with *Netcode for GameObjects*, future work should include more rigorous stress testing with larger numbers of annotations and users. This would help evaluate bandwidth usage, latency, synchronization robustness, and the limits of the current host-client architecture.
- **Heterogeneous client support:** since the current implementation focuses mainly on VR, future work could extend the system to desktop or web clients. This would allow a non-VR user to act as a facilitator or leader, managing the scene, guiding VR users, and handling more complex data operations.

In conclusion, while the current functionalities reflect its status as a prototype, the software architecture and ergonomic choices provide a scalable basis for future research into heterogeneous collaborative virtual environments.

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Code Availability

The source code of the prototype is available on GitLab at: https://git.unistra.fr/abates/mapcesium

Chapter 5

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